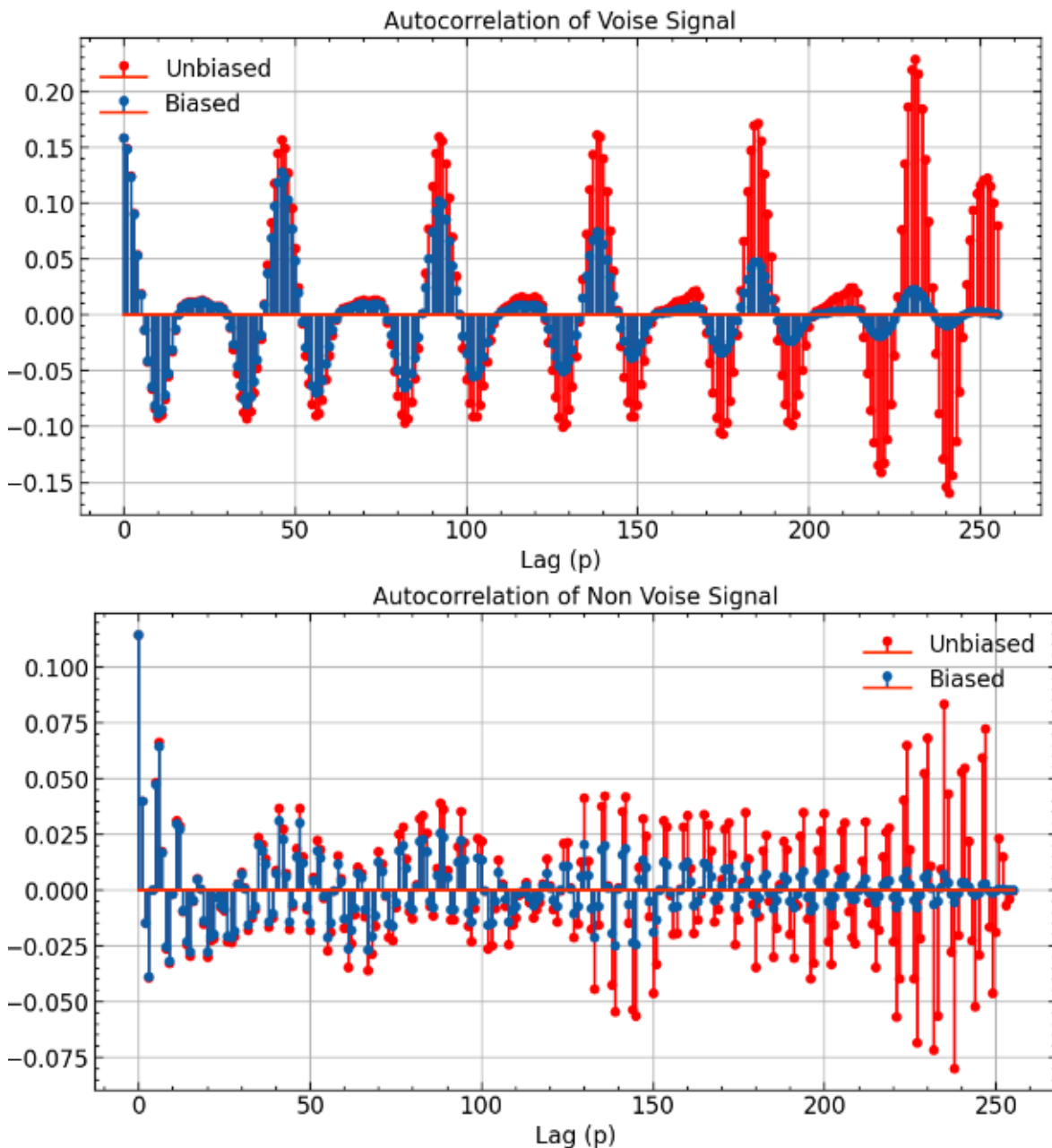


# MSA Lab : Graphs file

The graphs displayed will focus on the voiced and unvoiced signal. To remind our goal is to : make a coding and decoding chain with human voice using linear predictive coding.

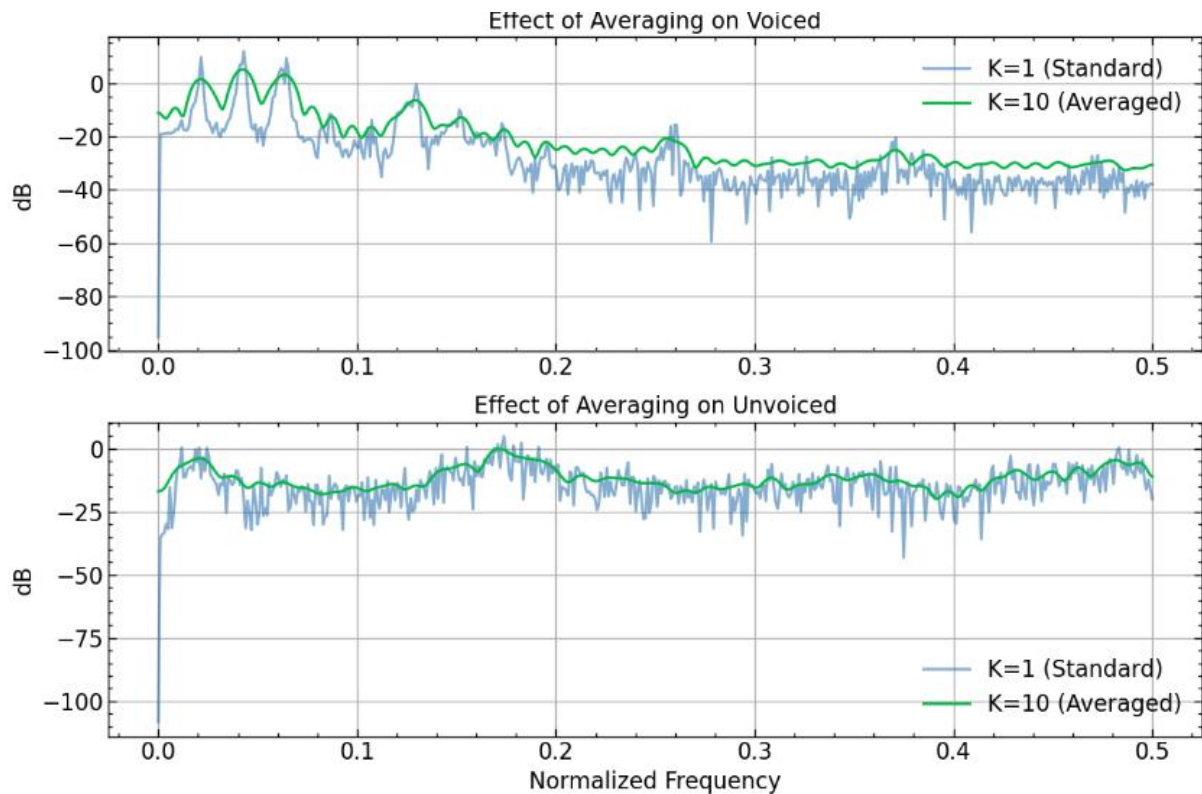
## I – Autocorrelation of Voice and Unvoiced signals



The autocorrelation of the white noise (close to the behavior of an unvoiced sound) is ideally 0 when  $p$  is different of 0. That's why trying to unbiased it makes it diverge; it exacerbates that error.

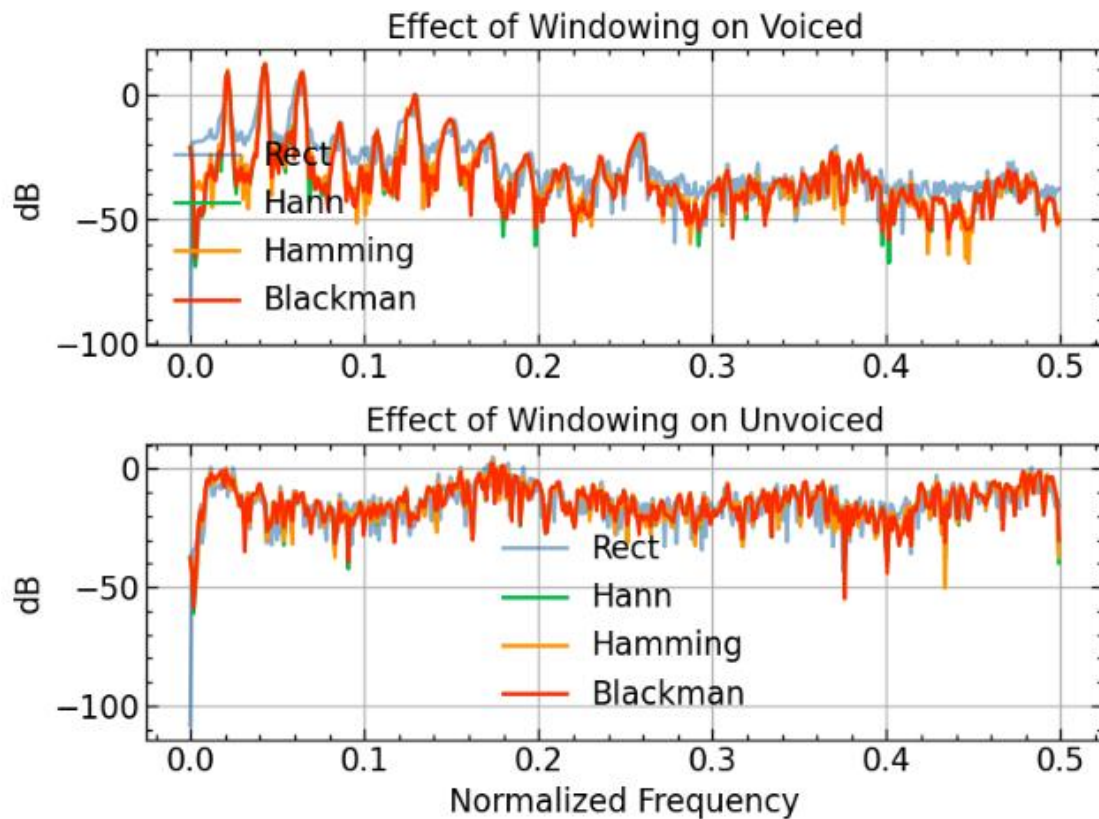
The voiced or unvoiced character is characterized by the quasi-periodic or non-quasi-periodic aspect of the signal. It can be seen by comparing the amplitude of the second maximum of the autocorrelation function to the maximum of the autocorrelation function.

## II – PSD of Voice and Unvoiced signals



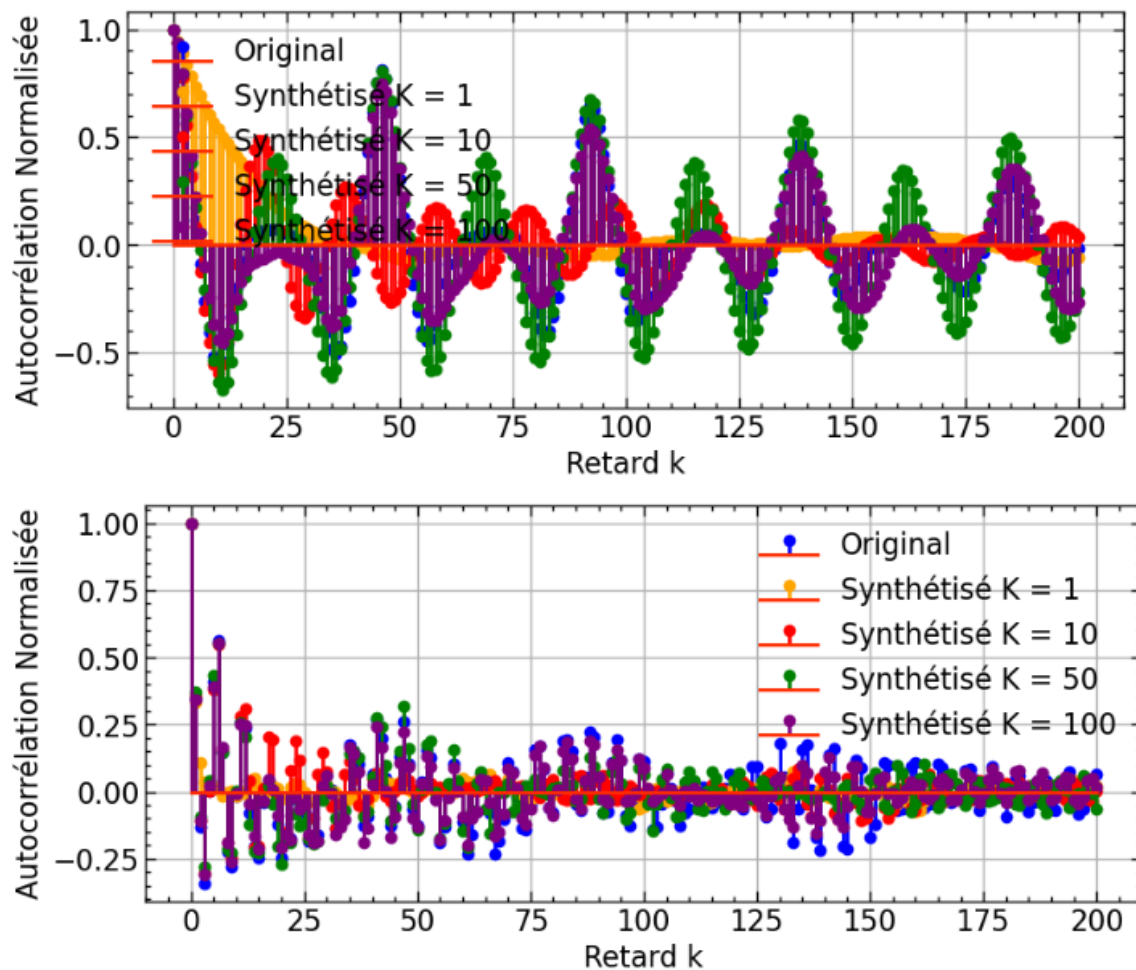
As said earlier, the unvoiced signal is close to the behavior of a white noise even PSD wise. It's constant compared to the voice one with 3 distinct pics at the beginning. The K has the easing expected.

### III – Windowing on Voiced and Unvoiced PSD



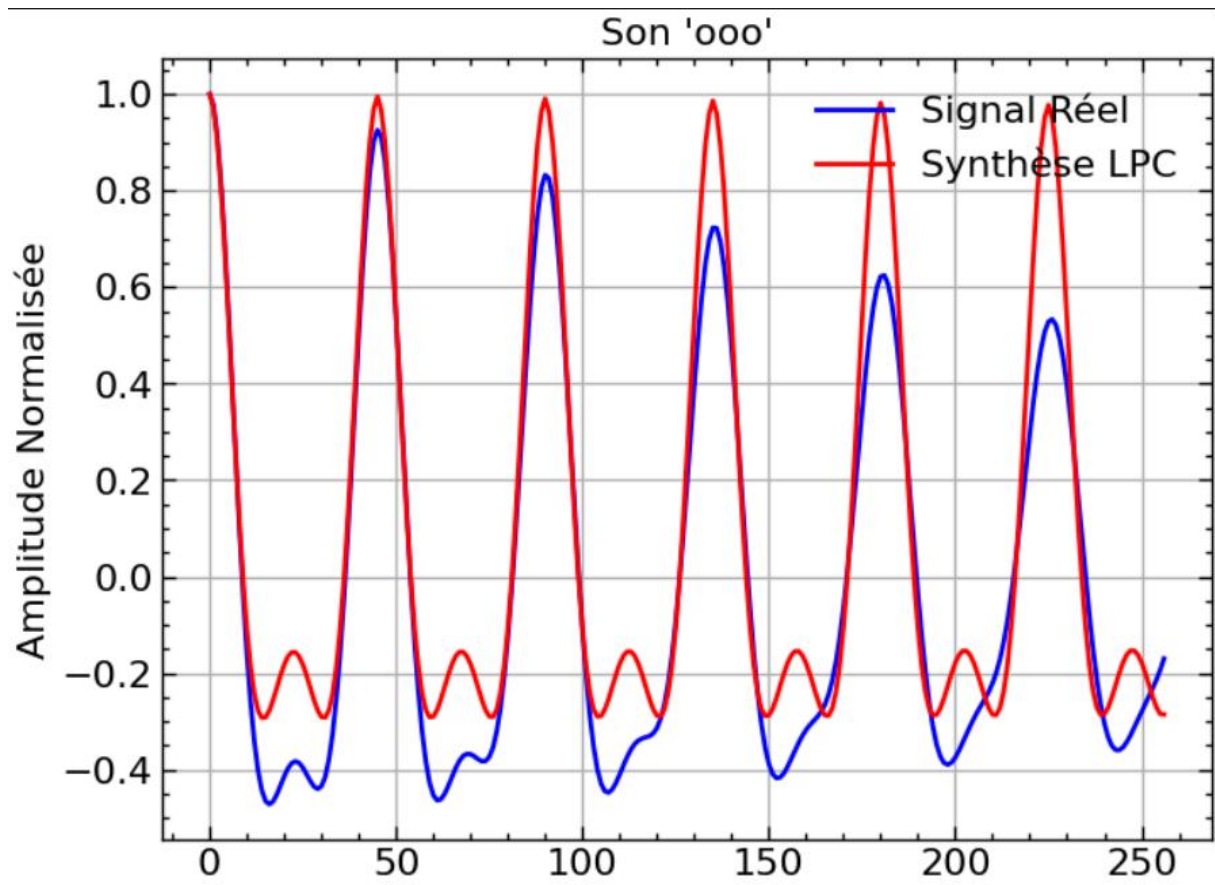
We can see the effect of windowing on voiced and unvoiced PSD. Windowing lowered the noise for the voiced signal so it can be useful for a better result at the end of the chain.

#### IV – Autocorrelation of synthesized signals using coefficients determined by the Yule-Walker equation



For both voiced and unvoiced signals, we observe that for  $K = 10$ , the autocorrelation aligns sufficiently with the original signal to be useful. For a higher  $K$  value, the error continues to decrease, but less sharply.

## V - Autocorrelation of a voiced signal 'ooo' and its synthesis



The autocorrelation of the original signal is superimposed on that of the synthesized signal. Comparing the original signal and the signal resynthesized from the compressed parameters reveals a near-perfect overlap of autocorrelations. This validates the entire chain: the model has captured both the necessary periodicity (Pitch) and spectral color (Filter).